Course Project Guidelines

Introduction to Deep Learning

Weight: 70% of the final course grade
Submission deadline: 18/05/2026 23:59 CET
Project presentations: To be confirmed in class
Project format: Individual or groups of up to two students

1 Introduction

This document describes the course project for Introduction to Deep Learning. The project constitutes the primary assessment component of the course and evaluates students' ability to design, analyse, and critically assess deep learning models in realistic prediction environments.

The project should be connected to one or more of the main topics covered in the course, including traditional computer vision, convolutional neural networks, image classification, transfer learning, data augmentation, recurrent neural networks (LSTM), traditional natural language processing, word embeddings, self-attention, and Transformer architectures.

The goal of the project is not simply to obtain good predictive performance, but to demonstrate a rigorous evaluation of the model, the methodology, and the behaviour of the selected approach. Students are expected to analyse not only when a model works, but also when it fails and why.

2 Project Types

Students may choose one of two project formats:

- Applied Project
- Research Project

Both formats require a written report and an oral presentation.

3 Applied Project

The applied project consists of building and evaluating a deep learning model on a dataset of your choice. The project may focus on computer vision or natural language processing with deep learning methods.

The goal is not simply to train a model, but to demonstrate a careful modelling process, a sound experimental design, and a critical evaluation of the results.

Students must submit:

- a written report in PDF format,
- the code used for the experiments,
- the dataset or instructions to obtain the dataset.

The report must clearly describe:

- the prediction task and dataset,
- the preprocessing steps,
- the modelling choices and architecture,
- the experimental design,
- evaluation and interpretation of model behaviour,
- failure cases and limitations,
- robustness and reliability of the results.

The project will not be assessed on the use of large or complex models, but on the ability to design, evaluate, and validate deep learning models in realistic prediction settings.

For projects focused on natural language processing, students may use traditional NLP methods, word embeddings, recurrent neural networks, attention mechanisms, or Transformer architectures covered in the course. However, the use of large language models, including commercial APIs or pretrained generative LLMs, is not permitted for NLP applications. The objective is for students to implement, train, adapt, or analyse models within the scope of the course rather than rely on external large-scale language models.

For projects focused on computer vision, students may use any suitable computer vision approach, including traditional image-processing methods, convolutional neural networks, pretrained vision models, transfer learning, data augmentation, or other relevant architectures. The chosen approach must be clearly justified and critically evaluated.

4 Research Project

The research project is a descriptive research report on a topic related to deep learning. The goal is to analyse and explain an existing research direction rather than to solve a new problem.

Students must:

- choose a clearly defined research topic,
- review the relevant literature,
- describe the current state of the art,
- analyse how existing approaches compare and why,
- discuss strengths, limitations, and open problems.

Where possible, small experiments are encouraged to test or illustrate hypotheses discussed in the report.

The report must:

- cite the relevant literature correctly,
- explain the problem and the proposed solutions clearly,
- analyse how different approaches compare,
- discuss what remains difficult or unresolved in the selected topic,
- provide a reasoned conclusion on which research directions appear most promising for a development team attempting to deploy such solutions in practice.

The objective is to demonstrate understanding, synthesis of the literature, and critical analysis.

5 Specific Deep Learning Deliverables

In addition to the general project requirements, the project must demonstrate a clear connection to the specific content of the Introduction to Deep Learning course.

Applied projects must include at least two of the following components:

- an analysis of learned representations, such as CNN feature maps, learned filters, word embeddings, attention weights, or intermediate model activations,
- a comparison between a traditional baseline and a deep learning model,
- an ablation study showing the effect of one deep learning component, such as batch normalization, data augmentation, transfer learning, word embeddings, positional encoding, or self-attention,
- an error analysis identifying examples where the deep learning model succeeds or fails, and explaining these cases in relation to the model architecture,
- a discussion of representation learning, explaining what the model learns automatically from data compared with manually designed features,
- for Transformer-based projects, an explanation of the role of self-attention and positional encoding in the selected model,
- for CNN-based projects, an explanation of the role of convolution, pooling, batch normalization, and learned filters in the selected model.

Research projects must include a clear explanation of how the selected topic relates to deep learning methods covered in the course and must critically discuss limitations, unresolved challenges, and practical difficulties.

6 Failure Analysis and Open Limitations

All projects must include a dedicated section on failure analysis. Students should not only report when the model performs well, but also identify and discuss cases where the model fails.

This section should include:

- examples of incorrect or unstable predictions,
- a discussion of possible reasons for these failures,
- analysis of whether the failures are related to the data, the model architecture, the training procedure, or the evaluation setting,
- a comparison between the failures of the deep learning model and those of a simpler baseline, where applicable,
- a discussion of what remains difficult or unresolved in the selected task, even after applying deep learning methods.

Students should use this section to reflect critically on the limitations of their approach and to explain what additional experiments, data, or modelling changes would be needed to better understand or improve the system.

7 Project Format and Submission

7.1 Written Report

All projects must include a written report in PDF format.

Typical structure:

- introduction and problem description,
- methodology or literature review,
- experimental or analytical section,
- failure analysis and limitations,
- discussion and conclusions.

Reports should be clear, structured, and well written. Figures and tables should be used where appropriate.

For applied projects, the report should explain the dataset, preprocessing, model architecture, training procedure, evaluation metrics, results, failure cases, and limitations.

For research projects, the report should explain the topic, relevant literature, comparison of approaches, limitations, open problems, and practical implications.

7.2 Code and Data

Applied projects must include the code used to run the experiments, as well as the dataset or instructions required to obtain it.

The submission should allow the instructor to inspect the modelling process and understand how results were obtained.

The code should be clearly structured and documented. External code, pretrained models, or online tutorials may be used only if they are clearly acknowledged and the student contribution is clearly explained.

For NLP projects, students must not use LLM-generated outputs as the main model, main experimental system, or main source of results. Any use of external tools must be clearly disclosed and must not replace the student's own implementation and analysis.

8 Grading Criteria

The project is graded out of 100 points.

Problem Definition and Conceptual Clarity: 15 points

- Clarity of the prediction problem or research question
- Understanding of the modelling challenge
- Connection with the course topics
- Quality of the conceptual framing

Experimental Design and Methodology: 20 points

- Appropriateness of modelling choices
- Sound experimental setup
- Proper use of training, validation, and testing procedures
- Reproducibility of results

Deep Learning Analysis: 20 points

- Clear explanation of the deep learning components used
- Analysis of learned representations or internal model behaviour
- Comparison with traditional or simpler baselines
- Critical discussion of why the deep learning approach is appropriate

Model Evaluation, Failure Analysis, and Validation: 20 points

- Quality of evaluation metrics
- Robustness checks and stress testing
- Detailed analysis of failure cases
- Discussion of limitations and unresolved challenges
- Ability to identify instability, bias, overfitting, or poor generalisation

Technical Quality and Implementation: 10 points

- Code clarity and structure
- Correctness of implementation
- Transparency of experimental pipeline

Presentation and Communication: 15 points

- Respect of presentation time
- Clarity and organisation of slides
- Ability to answer questions
- Critical reflection on results and limitations

The course project is intended to develop the ability to critically evaluate deep learning models beyond simple predictive accuracy. Students are encouraged to focus on careful experimental design, transparent analysis, and thoughtful interpretation of model behaviour.

A strong project does not need to use the largest model or the most complex architecture. A clear, well-designed, and well-evaluated project is preferable to a complex project with unclear methodology or unsupported conclusions.